

```
In [1]: import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn import metrics
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [3]: data = pd.read_csv("CH.SC.U4CSE24156_EX-7.6_2(Decision tree).csv")
data.head()
```

```
Out[3]:
```

	Weather	Go_Out
0	Sunny	Go Out
1	Sunny	Go Out
2	Sunny	Go Out
3	Sunny	Go Out
4	Sunny	Go Out

```
In [5]: data.shape
```

```
Out[5]: (15, 2)
```

```
In [7]: data.isnull().sum()
```

```
Out[7]: Weather    0
Go_Out    0
dtype: int64
```

```
In [9]: le_weather = LabelEncoder()
le_goout = LabelEncoder()

data["Weather"] = le_weather.fit_transform(data["Weather"])
data["Go_Out"] = le_goout.fit_transform(data["Go_Out"])

data.head()
```

```
Out[9]:
```

	Weather	Go_Out
0	2	0
1	2	0
2	2	0
3	2	0
4	2	0

```
In [11]: X = data[["Weather"]]
y = data["Go_Out"]
```

```
print(X.head())
print(y.head())
```

```
      Weather
0         2
1         2
2         2
3         2
4         2
0         0
1         0
2         0
3         0
4         0
Name: Go_Out, dtype: int32
```

```
In [13]: X_train, X_test, y_train, y_test = train_test_split(
        X,
        y,
        test_size=0.2,
        random_state=5
    )

    print(X_train.shape)
    print(y_train.shape)
    print(X_test.shape)
    print(y_test.shape)
```

```
(12, 1)
(12,)
(3, 1)
(3,)
```

```
In [15]: model = DecisionTreeClassifier(
        criterion="entropy",
        random_state=5
    )

    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    print(y_pred)
```

```
[1 0 1]
```

```
In [17]: conf_mat = metrics.confusion_matrix(y_test, y_pred)

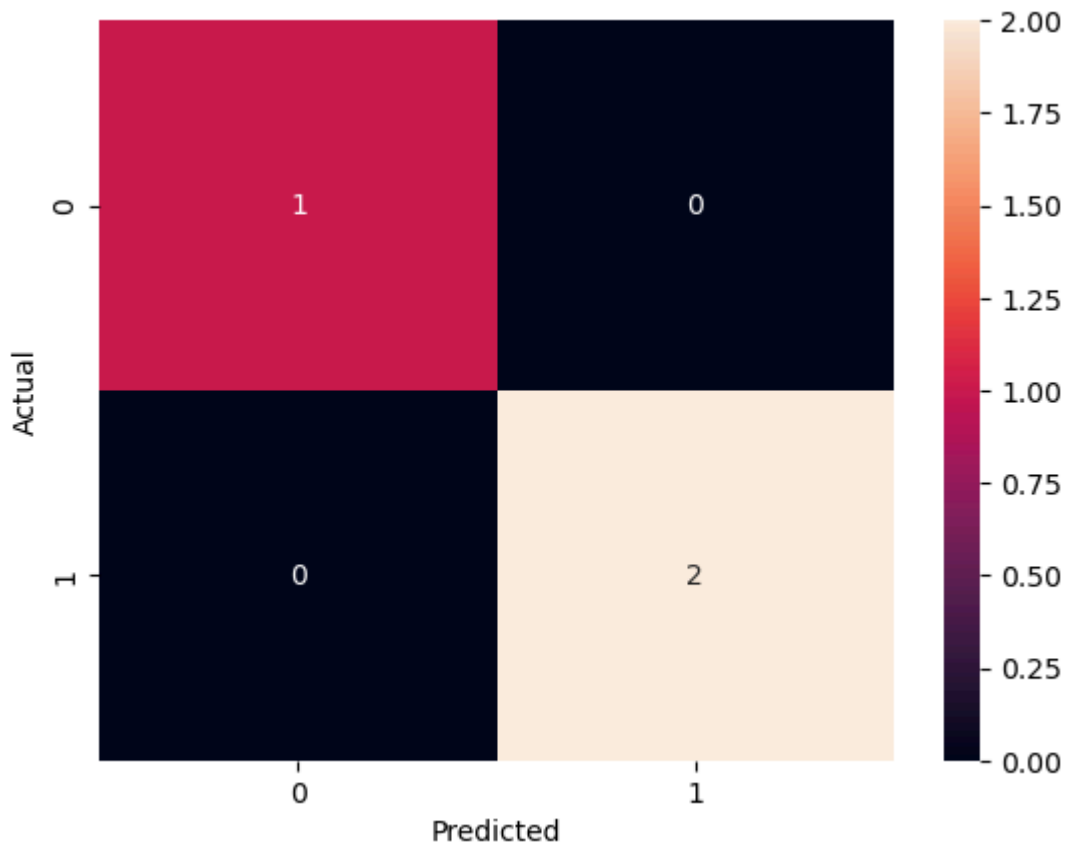
    print("Confusion Matrix:")
    print(conf_mat)

    accuracy_score = metrics.accuracy_score(y_test, y_pred)

    print("Accuracy Score:", accuracy_score)
    print("Accuracy in Percentage:", int(accuracy_score * 100), "%")
```

```
Confusion Matrix:
[[1 0]
 [0 2]]
Accuracy Score: 1.0
Accuracy in Percentage: 100 %
```

```
In [19]: conf_mat = pd.crosstab(  
    y_test,  
    y_pred,  
    rownames=["Actual"],  
    colnames=["Predicted"]  
)  
  
sns.heatmap(  
    conf_mat,  
    annot=True  
)  
  
plt.show()
```



```
In [ ]:
```